

Mod List

Minecraft 1.21.1 — NeoForge

Complete mod documentation

Version 3.0.5

Wave -1 — Prerequisites

JDK 21

Tool	Role
Eclipse Temurin JDK 21 (LTS)	Java runtime and development kit for Minecraft 1.21.1

Install the **JDK** (not just JRE) — the JDK includes `javac` and `jlink` needed by NeoForge. Minimum version 21.0.0, any LTS build. On Windows, use the MSI installer and ensure `JAVA_HOME` is set to the install path (e.g., `C:\Program Files\Eclipse Adoptium\jdk-21.0.6.13-hotspot`). On Linux/macOS, use your package manager (`apt install temurin-21-jdk`, `brew install temurin21`).

Verify installation:

```
java -version
# Expected: openjdk version "21.0.x" ...
```

Performance Tuning

Recommended JVM arguments (for GDLauncher → right-click instance → Settings → Java & Memory → JVM Arguments):

```
-XX:+UseZGC -XX:+ZGenerational -XX:+AlwaysPreTouch -XX:+PerfDisableSharedMem -
XX:SoftMaxHeapSize=12G -Xms8G -Xmx8G
```

System RAM	Recommended allocation
16 GB	<code>-Xms10G -Xmx10G</code>
32 GB	<code>-Xms22G -Xmx22G</code>
64 GB	<code>-Xms48G -Xmx48G</code>

Note: ZGC (Z Garbage Collector) with generational mode is the recommended GC for Java 21 with NeoForge. It provides consistently low latency (sub-millisecond pause times) and handles the large heap sizes common with modded Minecraft better than Shenandoah or G1GC. The Adoptium Temurin JDK 21 includes ZGC — no special Java build needed. `-XX:+AlwaysPreTouch` pre-initializes all memory pages at startup, preventing runtime page-fault stalls. `-XX:+PerfDisableSharedMem` disables synchronous `hsperfdata` file writes, removing a minor source of I/O-induced latency. `-XX:SoftMaxHeapSize=12G` tells ZGC to target ~12G heap usage and GC more aggressively to stay near that level, while allowing bursts above it up to `-Xmx`. For this to take effect, `-Xmx` must be set higher than `-Xms`. The baseline uses equal values for simplicity; on a 64 GB machine you may want e.g. `-Xms12G -Xmx24G` instead to let `SoftMaxHeapSize` work.

GDLauncher

Tool	Role
GDLauncher	Mod manager and launcher for the modpack

GDLauncher manages mod and modpack installation, updates, and launching for this modpack. It supports both **CurseForge** and **Modrinth** as mod sources in one launcher, handles NeoForge installation automatically, manages Java versions per instance, and lets you install mods with one click. Free, with no premium tier.

Setup:

1. Download and install [GDLauncher](#) for your OS.
2. Create a new instance: click + on the Library page → select Minecraft **1.21.1** and modloader **NeoForge** (latest recommended for 1.21.1).
3. GDLauncher downloads NeoForge and the correct Java version automatically.
4. Launch the instance once to generate the `mods/` directory and configs.
5. Verify the main menu shows “NeoForge X.X.X” in the bottom-left corner.

From there, add mods through the instance’s **Addons** tab from either CurseForge or Modrinth, or by dropping `.jar` files into the instance’s `mods/` folder. All mod links in this document link to their CurseForge pages — GDLauncher can install from CurseForge directly via the Addons tab.

For texture packs, drop `.zip` files into the instance’s `resourcepacks/` folder. For shaderpacks, drop `.zip` files into the instance’s `shaderpacks/` folder. Access the instance folder by right-clicking the instance and selecting **Open Folder**.

Minecraft 1.21.1

Install the **Minecraft 1.21.1** vanilla client from the official launcher:

1. Open the [Minecraft Launcher](#).
2. Create a new installation with version **release 1.21.1**.
3. Run it once to generate the vanilla `versions/` and `libraries/` directories, then close.
4. Locate your Minecraft directory:
 - Windows: `%APPDATA%\minecraft`
 - Linux: `~/.minecraft`
 - macOS: `~/Library/Application Support/minecraft`

NeoForge

Tool	Role
NeoForge	Mod loader for Minecraft 1.21.1

NeoForge is the mod loader powering this modpack. If you’re using **GDLauncher**, it installs NeoForge automatically when you create a NeoForge instance (see [GDLauncher](#) above) — skip to step 5 to verify.

If installing manually:

1. Download the **NeoForge installer** for Minecraft 1.21.1 from [neoforged.net](#).
2. Run the installer: `java -jar neoforge-1.21.1-<version>-installer.jar`.
3. Select “**Install client**” and point it to your Minecraft directory (auto-detected on most systems).
4. Launch the new **neoforge** profile in the Minecraft Launcher once to generate the `mods/` directory and NeoForge configs.
5. Verify the main menu shows “NeoForge X.X.X” in the bottom-left corner.

Important: Do not add any mod `.jar` files yet. Wave -1 is complete when the vanilla 1.21.1 client runs under NeoForge with no mods installed.

Wave 0 — Foundation

Dependencies

Mod	Role
Cloth Config	Config screen library
Architectury API	Cross-loader compatibility
YACL	Config library
Curios API	Accessory slots (replaces Trinkets)
CreativeCore	Library (AmbientSounds)
OctoLib	Library (Immersive UI)
Fragmentum	Library (Loot Journal)
M.R.U	Library (Sounds)
YUNG's API	Library (Traveler's Titles, YUNG's Better mods)
Not Enough Animations	First-person animation library (First Person Model dependency)
Balm	Multi-loader library (Inventory Essentials)
Kotlin for Forge	Kotlin stdlib — required by Kotlin-based mods (e.g. TFMG Energy Converter)

Note: GDLauncher installs dependencies **automatically** when you install a mod through its Addons tab. You do not need to manually download or install any dependency entries in this document — just install the mods themselves and GDLauncher pulls in what's needed.

Performance & Rendering

Mod	Role
Sodium	Rendering engine
Iris	Shader loader (v1.8.14, NeoForge native)
Lithium	Server-side game logic optimization
Distant Horizons	LOD rendering
ModernFix	All-in-one perf + bug fixes
ImmediatelyFast	Immediate-mode rendering
Entity Culling	Hide off-screen entities
Dynamic FPS	Background FPS reduction
NoisiumForked	Worldgen optimization (active fork)
FerriteCore	Memory usage reduction

Visual & Client QoL

Mod	Role
LambDynamicLights	Dynamic lighting
BetterF3	Customizable debug HUD
Sound Physics Remastered	Realistic sound
AmbientSounds	Nature ambiance (birds, wind, water)

Mod	Role
The Immersive Music Mod	147 new biome/structure-based songs, vanilla silence gaps preserved
Immersive UI	Animated UI, smooth hotbar, item particles
Status Effect Bars Reforged	Duration bars on status effect icons in HUD and inventory
Traveler's Titles	RPG-style titles entering biomes/dimensions
Better Modlist	Enhanced mod list screen with badges/categories
Better Third Person	Independent 360° camera rotation in third-person view
Tooltip Overhaul	Modern, sharp tooltip rendering with equipment compare
First Person Model	Full body rendering in first-person view (requires Not Enough Animations)
3D Skin Layers	Renders player skin layers (jackets, hats) with 3D depth
Presence Footsteps	Block-aware footstep sounds for every surface type
Particular Reforged	Ambient particles: 3D water splashes, fireflies, falling leaves, cave dust, waterfall cascades
Wakes Reforged	Realistic water wakes and splashes when traveling on water
Sounds	170+ new sound effects for UIs, items, blocks, actions
Better Days	Customizable day/night cycle length, enhanced sleep
Better Safe Bed	Prevents mobs that can't reach you from blocking sleep — chill-rule essential
Beautiful Enchanted Books [Mod Edition]	Unique textures per enchantment book type — standalone mod, no resource pack needed
Overflowing Bars	Expanded health, armor, and toughness bars beyond vanilla limits
Cherished Worlds	Favorite/pin worlds to top of list, prevent accidental deletion
Day Counter	Day counter title overlay with milestone celebrations — appears at dawn each day
Eating Animation	First-person eating/drinking animation — food moves to mouth and shrinks
Camera Overhaul	Camera tilting, screen shakes, mouselook roll, idle sway — satisfying movement feedback
[DLS] Dark Loading Screen	Replaces the default bright-red loading screen with a dark version — easier on the eyes

Shaderpack

Shaderpack	Description
Complementary Unbound	Realistic visual style, Potato→Ultra profiles, block-specific effects

Settings for NVIDIA RTX 4080 SUPER (16GB VRAM, 4K) — target: stable 70-80 FPS:

Setting	Value
Visual Style	Unbound
Profile	High
RP Support	Integrated PBR+
Shadow Resolution	2048
Shadow Distance	12 chunks
Real-Time Shadows	High
Light Shaft Quality	Medium
Entity Shadows	OFF
Detail Quality	High
Cloud Quality	High
Water Reflection Quality	Medium (max)
Block Reflection Quality	Medium
Anti-Aliasing (FXAA)	ON
Edge Shadow (SSAO) Quality	Medium
Advanced Colored Lighting	8 chunks
Texture Filtering	8x
Bloom	OFF

General MC settings: Graphics **Fancy**, Render Distance **10 chunks**, Simulation Distance **8 chunks**, Biome Blend **2 blocks**, Mipmap Levels **4x**, Entity Distance **100%**, Fullscreen ON, VSync OFF, Max Framerate **120**. Always Defer Chunk Updates **Enabled**, Use No Error Context **Enabled**.

Distant Horizons handles far rendering — keep vanilla render distance low. DH LODs render through the shader pipeline; start LOD distance at **128** and push to **256** if the FPS budget allows. Monitor VRAM usage (shaders + DH + high-res texture packs can peak past 12GB at 4K). If you experience dips below 70 FPS, first reduce Shadow Resolution to **1024**, then lower Shadow Distance to **8 chunks**.

Inventory & UI

Mod	Role
JEI	Recipe viewer
Jade	Block info HUD
Jade Addons (Neo/Forge)	Extra Jade info panels
JourneyMap	Real-time minimap + full-screen world map, waypoints, entity radar
JourneyMap Integration	Shows Waystones as icons on JourneyMap
AppleSkin	Food hunger/saturation overlay
Mouse Tweaks	Inventory management shortcuts

Mod	Role
Inventory Essentials	Drag-to-transfer, single-item moves, sorting
Fancy Toasts	Beautiful animated advancement popups
Shulker Box Tooltip	Preview shulker box contents on hover
Item Borders	Colored borders on items based on rarity
Loot Journal	Animated item pickup notifications
Reliable Advancements	Overhauled advancements UI with editor/pan/zoom
Polymorph	Choose crafting result when recipes conflict

Storage & Travel

Mod	Role
Sophisticated Storage	Upgradable barrels/chests
Sophisticated Backpacks	Portable inventory
Sophisticated Storage Create Integration	Full-featured Sophisticated Storage on Create contraptions
Sophisticated Backpacks Create Integration	Full-featured Sophisticated Backpacks on Create contraptions
Waystones	Fast travel between waypoints
GraveStone Mod	Keep inventory on death

Infrastructure

Mod	Role
Chunk Loaders	Keep chunks loaded across dimensions — essential for Phase 2+ automation
SuperMartijn642's Config Lib	Config library (Chunk Loaders dependency)
SuperMartijn642's Core Lib	Core library (Chunk Loaders dependency)

Village & Trading

Mod	Role
Dynamic Villager Trades Neoforge 1.0.0	Dynamically changing villager trades — villagers specialize in what you buy from them, reducing the need for massive trading halls
Improved Village Placement	Forces villages to spawn on flat terrain — prevents cliff-side and underwater villages

Configuration — Village & Trading

Dynamic Villager Trades (`config/dynamicvillagertraders.properties`):

- `global_randomness = 0.7` (recommended, default 1.0) — Lower values make villagers settle on specific trades faster; higher values increase time to get desired trades. Multiplied by group randomness in the datapack.
- `refresh_delay = 1` (recommended, default 0) — Number of restocks between trade changes. `1` means trades change every other restock, giving you time to use existing trades before they rotate. Trades always change on level-up regardless.

- `no_book_duplicates = true` — Prevents librarians from selling the same enchant at different levels simultaneously.
- `enchant_repair_compat = false` — Adds Curse of Vanishing to every book sold by librarians (for Enchant & Repair compatibility). Keep disabled unless that mod is added.

Texture Packs

Pack	Role
Fresh Animations	Living entity animation overhaul — mobs blink, look around, show emotion
Alacrity	Full 32x conversion — RPG-themed textures, GUI overhaul, custom 3D models
Alacrity — Just Enough Items Support	Retextures JEI (+8 addons) to match Alacrity's RPG style
Alacrity — Shulker Box Tooltip Support	Retextures Shulker Box Tooltip previews to match Alacrity's RPG style

| [Enhanced Boss Bars](#) | Redesigned boss bars with themed designs for vanilla and modded bosses | | [RAY's 3D Rails](#) | 3D block & item textures for all rail types | | [Better Lanterns](#) | Enhanced 3D look for lanterns and chains | | [Smooth Font 32x](#) | Smooth 32x font replacement for cleaner text readability | | [Vervada's 3D Plants](#) | 3D models for almost all plants — more immersive flora |

Fresh Animations adds idle animations, directional looking, sleep/blink cycles, and emotional states to all vanilla mobs. It's a resource pack overlay (not a mod) — place it high in the pack order. Works on any version with no mod dependencies.

Alacrity is a full 32x conversion with an RPG-inspired art style — all blocks, items, GUI elements, and mobs are redesigned with a cohesive fantasy look. Draws from Warcraft, Elder Scrolls, and Gothic for its aesthetic. Custom 3D models (CEM) require a mod like Entity Model Features but the base pack works with any setup.

Alacrity Mod Support packs (by TheDatThai) extend Alacrity's RPG styling to mod UIs: JEI and Shulker Box Tooltip. Place these **below** the base Alacrity pack — putting them above corrupts JEI/Shulker graphics (mismatched textures).

Enhanced Boss Bars replaces boss bar textures with themed designs — covers vanilla bosses and has support for Twilight Forest bosses. For full compatibility with L_Ender's Cataclysm and The Aether (which use custom boss bar rendering), install the companion [Enhanced Boss Bars mod](#). Pure resource pack for vanilla/Twilight Forest bars.

RAY's 3D Rails replaces rail block and item textures with 3D models. Supports shaders since v3.0. Compatible with all rail types from Steam 'n' Rails and Create.

3D Ladders replaces the vanilla ladder model with a 3D version. Simple, lightweight, no conflicts.

Better Lanterns gives lanterns and chains an enhanced 3D look with connected chain models. Compatible with Arcane Lanterns and Thin Air mods.

Installation: Install via GDLauncher — drop `.zip` files into the instance's `resourcepacks/` folder (right-click instance → Open Folder). Load order (top = highest priority):

1. Fresh Animations
2. Alacrity
3. Alacrity — Just Enough Items Support

4. Alacrity — Shulker Box Tooltip Support
5. Enhanced Boss Bars
6. RAY's 3D Rails
7. 3D Ladders
8. Better Lanterns
9. Smooth Font 32x
10. Vervada's 3D Plants

Configuration — Wave 0

Waystones (`config/waystones-common.toml`):

- `inventoryButton = "any"` — Opens waystone menu from inventory without a keybind.
- `enableCosts = false` — Free teleportation.

Loot Journal (`config/obscuria/loot_journal-client.toml`):

- `trackItemPickups = true` — Logs all item pickups for per-session statistics.
- `trackXpPickups = true` — Logs all XP pickups for per-session statistics.

Better Days (`config/betterdays-common.toml`):

Time speed can be set via three modes (`speedMethod`):

Mode	How it works	Best for
<code>MINUTES</code> (default)	Set <code>daySpeedMinutes</code> and <code>nightSpeedMinutes</code> in real minutes	Simple: 10 min day / 10 min night
<code>RATIO</code>	Multiplier relative to vanilla (1.0 = vanilla 20min cycle)	Fine control: 0.5 = 40 min, 2.0 = 10 min
<code>SEASON</code>	Day length varies by season using <code>seasonDayMinutes</code> and <code>seasonLatitude</code>	Immersive — day shortens in winter, lengthens in summer

Recommended settings for this pack (`speedMethod = "MINUTES"`):

- `daySpeedMinutes = 20` — Days last 20 minutes (2x vanilla). Gives relaxed building time.
- `nightSpeedMinutes = 10` — Nights remain 10 minutes. Sleep acceleration keeps them short.
- `enableSleepFeature = true` — Smooth time acceleration while in bed.
- `dayStart = 0` — Dawn at time 0. Night starts at 13000 (default, no change needed).
- `nightStart = 13000` — Dusk starts at time 13000 (default).

Time effects — all disabled by default. Leave them off:

- `weatherEffect = "SLEEPING"` — Lets rain/storms finish faster when sleeping (safe — NeoForge handles this natively anyway).
- `potionEffect = "NEVER"` — Keep off. Speeding potion timers while sleeping is confusing.
- `hungerEffect = "NEVER"` — Keep off. Speeding hunger while sleeping is annoying.
- `blockEntityEffect = "NEVER"` — Keep off. Enabling this accelerates furnaces/hoppers during sleep, which can desync items and wastes fuel.
- `cropEffect = "NEVER"` — Keep off. Conflicts with Serene Seasons crop cycle timers.

Regarding the SEASON mode: Since this pack includes **Serene Seasons**, you may want to try `speedMethod = "SEASON"` (not "SEASONS" — the value is singular) for a more immersive cycle where summer days are long and winter days are short. However, this is experimental — Serene Seasons already handles seasonal crop growth and temperature; Better Days would only change day length visually. Test before committing.

The `seasonDayMinutes` and `seasonLatitude` fields exist in the default config but may not appear in the file until `speedMethod = "SEASON"` is set and the game reloads. If they're missing, add these lines manually under the `[time]` section:

```
speedMethod = "SEASON"
seasonDayMinutes = 20.0      # Total day+night length. 20 = vanilla.
seasonLatitude = 48.0        # Central Europe latitude. -90 (short days) to 90 (long
                              days). Serene Seasons detects this automatically.
```

ModernFix (configurable in-game via Mod List → ModernFix → Config, or `config/modernfix-mixins.properties`):

- `mixin.perf.dynamic_resources = true` — Load block/item models lazily instead of all at startup. **Biggest single launch-time speedup** in the pack. Reduces memory usage too. Disable if you see missing textures or broken models.
-

Wave 1 — Tech (Create + Mekanism + AE2)

The classic tech trio. Create handles mechanical automation, Mekanism handles industrial processing, AE2 handles digital storage and autocrafting.

Core Tech

Mod	Role
Create	Mechanical automation — rotational power, water wheels, windmills, gear trains, conveyor belts, deployers, sequenced crafters, trains
Mekanism	Industrial processing — ore multiplication (2x→5x), digital miner, superdense energy storage, MekaSuit (flight, invulnerability)
Mekanism Generators	Power generation — heat, wind, solar, gas-burning, bio generators, fission reactor, fusion reactor, turbine
AE2 19.2.17	Digital storage — ME drives, autocrafting, P2P tunnels, spatial storage. The storage backbone
Applied Mekanistics	AE2-Mekanism integration — autocrafting for chemical/gas/fission/fusion recipes, ME chemical storage, P2P chemical tunnels

Note: Mekanism v10 for 1.21.1 NeoForge is split into separate modules. **Mekanism** (core) provides machines, processing, cables, and armor. **Mekanism Generators** provides all power generation. Both must be installed.

Create Addons

Mod	Role
Create Crafts & Additions 1.6.0	Electric motors, alternators, rolling mill, wires/rods, silver/electrum — bridges kinetic SU and Forge Energy (FE)
Create: Power Grid 0.5.5.1	Physics-based electricity simulation — Ohm's law, AC/DC, generators, motors, cables, FE conversion. Separates Create power from SU
Steam 'n' Rails NeoForge 0.2.1	Expanded train system — new tracks (spruce, monorail), semaphores, conductor mob, coupling/decoupling blocks
Create: The Factory Must Grow (TFMG)	Heavy engineering & oil — crude oil drilling, distillation, diesel/gasoline/LPG engines, steel, aluminum, electricity with voltage
Mekanism TFMG Compatibility	85+ recipe bridges — deduplicates lead/steel/sulfur, integrates aluminum/plastic into Mekanism chains
TFMG Energy Converter 26.03	Bidirectional TFMG electricity ↔ FE/AE2 bridge — lets TFMG diesel engines/generators power non-TFMG machines (Mekanism, AE2). Requires Kotlin for Forge
Create: Gunsmithing	Steampunk firearms integrated with Create — craftable using Create machines. Animated guns, attachments, ammo types. Optional Better Combat compatibility

Mod	Role
Create Ornithopter Glider	Mechanical flapping glider with boost-assisted flight — integrates with Create rotational power
Create: Train Utilities 3.0.3	Sliding doors with realistic sounds, platform blocks, station building blocks
Create Railways Navigator	Display boards, route navigation, new schedule entries — search routes and manage train networks
Create: Interiors 0.6.1	Colored chairs with armrest toggles, tray tables, kelp seats — functional furniture for train cars
Create Train Parts 0.4.1	Crossing gates, sliding windows, train steps, slides — decorative and functional blocks for trains and stations
Create: Threaded Trains	Performance — runs train pathfinding on a separate thread, prevents tick lag on complex rail networks

Create: Gunsmithing Dependencies

The following library mods are required by Create: Gunsmithing:

Mod	Role
[NTGL] NukaTeam's Gun Lib	Gun framework — animated weapons, ammo, attachments, gun packs. Required by Create: Gunsmithing
GeckoLib	3D animation library — required by NTGL
Framework	Utility library — required by NTGL

Mekanism Addons

Mod	Role
Advanced Finders	Ore detection devices — directional indicators for nearby ore veins, synchronized multiplayer support
Advanced Chimneys	Smoke ventilation systems — redirect factory emissions outdoors, customizable appearance and behavior

Dependencies: Both Advanced Finders and Advanced Chimneys require [ForgeEndertech](#) library.

How They Work Together

Layer	Mod	
Power generation	Create (water wheels, windmills, steam) + Mekanism (advanced generators, fusion reactor)	
Ore processing	Mekanism — up to 5x ore multiplication with chemical processing	
Factory automation	Create — conveyor belts, deployers, sequenced crafters	
Storage	AE2 — ME drives, storage busses, crafting CPUs	
Transport	Create trains + AE2 P2P tunnels	
Endgame	Mekanism fusion reactor + MekaSuit + AE2 autocrafting	### Ores

Mod	Role
Create (built-in)	Zinc, brass, copper
Mekanism (built-in)	Osmium, tin, lead, uranium, fluorite

Configuration — Wave 1

Mekanism (`config/Mekanism/general.toml`):

- `allowChunkloading = false` — Disable Mekanism’s built-in chunk loading. Prevents conflict with the dedicated Chunk Loaders mod and keeps chunk management unified.

Ore generation: All default rates. Osmium at iron-like frequency, tin/lead moderate, uranium/fluorite rarer. Default distribution aligns with the pack’s progression curve.

Interdependency notes for Wave 1:

- Mekanism’s universal cables can power AE2 controllers via the Energy Acceptor. No adapter mod needed.
- **Create Crafts & Additions** electric motors and alternators create a direct kinetic↔FE bridge, replacing the Create Alternator workaround. Silver and electrum add new material tiers that integrate with Mekanism’s ore processing line.
- **Steam ‘n’ Rails** expands Create’s train system with new tracks, semaphores, conductor mobs, and coupling blocks. Essential for moving resources between MineColonies, Mekanism factories, and AE2 storage across a sprawling base.
- **Create train ecosystem:** SnR provides track types (monorail, wooden, wide gauge), semaphores, and the conductor mob. **Train Utilities** adds station architecture (sliding doors, platform blocks). **Railways Navigator** adds display boards and route management. **Interiors** makes train cars usable with seats and tray tables. **Train Parts** adds crossing gates, windows, and steps for detailing. **Threaded Trains** keeps the whole network performant by running pathfinding on a separate thread. Together they cover tracks → stations → navigation → interiors → detailing → performance without overlap.
- Mekanism’s 5x ore processing (Chemical Dissolution Chamber → Washer → Crystallizer) outclasses Create’s Crushing Wheels (1.5x) and Millstone (1x) for ore multiplication. This is intentional — Create handles mechanical automation and component crafting, Mekanism handles industrial processing. Neither obsoletes the other.
- AE2’s Storage Bus on a Sophisticated Storage barrel makes early-mid game storage migration seamless. Place a Storage Bus on your barrel wall, configure it to show contents in the ME terminal, and transition gradually to full digital storage.
- **Sophisticated Storage Create Integration** and **Sophisticated Backpacks Create Integration** (both by P3pp3rF1y) enable full backpack and barrel functionality on Create contraptions — upgrades, automation features (pickup, magnet, feeding), and inventory settings all work while the contraption is in motion. Without these addons, Sophisticated inventories on contraptions are limited to basic item transport.
- **Applied Mekanistics** bridges AE2 autocrafting with Mekanism’s chemical processing — ME Chemical Output Buses, ME Chemical Import Buses, and P2P tunnels for chemicals/gases enable autocrafting of Mekanism recipes (including fission/fusion products) directly through the AE2 crafting CPU. Without this mod, Mekanism chemical steps are manual bottlenecks.
- **Create: The Factory Must Grow** adds oil drilling, distillation towers, blast furnaces, and diesel engines — expanding Create’s mechanical toolkit into heavy industry. Its steel replaces Mekanism’s steel (TFMG’s blast furnace is the intended steel source; Mekanism’s steel recipe is disabled by the Mekanism TFMG Compat mod). TFMG aluminum and plastic feed into Mekanism’s advanced processing chains.

- **Mekanism TFMG Compatibility** resolves 85+ recipe overlaps between TFMG and Mekanism — lead, sulfur, and steel are unified under TFMG's production lines, while Mekanism retains its chemical processing advantage. Both JEI item blacklist config and per-recipe toggles are available for pack tuning.
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Wave 1.5 — Colony Management

MineColonies provides autonomous NPC workers that mine, farm, craft, and guard your base.

Colony Automation

Mod	Role
MineColonies	Autonomous NPC colony — builders, miners, farmers, crafters, guards, research tree
Structurize	Structure building/placement library (MineColonies dependency)
MultiPiston	Multi-block piston library (MineColonies dependency)
BlockUI	UI library (MineColonies dependency)
Domum Ornamentum	Decorative block library (MineColonies dependency)

Interdependency notes for Wave 1.5

- MineColonies guards (deployed via the Town Hall’s “Guard Tower” building) patrol your base perimeter, reducing the need to secure every wall gap manually. Their effectiveness scales with the Guard Tower level and the colony’s skill research. This creates a natural progression — automated base defense grows as your colony does.
 - MineColonies Colony Research milestones signal Phase 2 progression readiness. For example, the “Industrial Revolution” research (Colony level 3, 5000 research points) serves as a soft gate — by the time you reach it, you have the colony infrastructure to support Mekanism and AE2. Hard recipe gating (e.g., Atomic Alloy locked behind research) requires a data pack or scripting mod and is not implemented in this pack — the progression pillars in AGENTS.md define the intent, and playtesting determines if soft gating is sufficient.
 - Chunk Loaders (Wave 0) are essential for MineColonies — they keep workers active while you explore other dimensions. Place at least one at your colony after setting up your base.
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Wave 2 — Exploration (World & Dimensions)

Overworld Worldgen

Mod	Role
Terralith	Overhauled Overworld biomes
Tectonic	Better terrain shape — deeper valleys, taller mountains, cliffs

Environment

Mod	Role
Serene Seasons	Seasonal foliage colors, temperature shifts, crop growth cycles

Serene Seasons adds four seasons (Spring, Summer, Autumn, Winter) with visual foliage changes, temperature shifts, and seasonal crop growth modifiers. Crop impact is enabled — planting in the wrong season reduces yield, encouraging greenhouse planning. Temperature damage is **disabled** — no freezing or heat damage, preserving the chill rule.

Underground & Ocean

Mod	Role
TerraBlender (NeoForge) — REQUIRED	Worldgen biome library — required by Darker Depths 2.1+ for biome placement
Darker Depths	Underground expansion — 3 new cave biomes, mobs, Forsaken Bronze, Living Crystals
Upgrade Aquatic	Ocean expansion — Thrasher mob, Ocean Ravines, Prismatic Coral, renewable sand

Interdependency note — Darker Depths + TerraBlender: Darker Depths 2.1+ requires a worldgen API to register its cave biomes. Either **TerraBlender** (NeoForge) or **Biolith** must be installed. TerraBlender NeoForge is the recommended choice (available for 1.21.1 on both CurseForge and Modrinth). Without it, Darker Depths crashes on startup with **Missing Worldgen API! Install TerraBlender or Biolith.**

Interdependency note — Darker Depths + YUNG's Better Caves: Darker Depths adds its own cave biomes (Molten Cavern, Sandy Catacombs, Crystal Cavern) that generate alongside YUNG's Better Caves' enlarged cave systems. Darker Depths biomes replace vanilla cave sections with themed variants, while YUNG's Better Caves changes the shape and connectivity of caves — they layer without direct conflict.

Interdependency note — Upgrade Aquatic + YUNG's Better Ocean Monuments: Upgrade Aquatic adds ocean mobs, coral variants, and ocean-floor features. It does not modify Ocean Monuments — YUNG's Better Ocean Monuments replaces monument structure generation without conflict. Upgrade Aquatic items (thrasher teeth, prismatic coral) integrate naturally with Ocean's Delight (Wave 4) for seafood recipes.

Structures

Mod	Role
YUNG's Better Dungeons	Overhauled dungeons
YUNG's Better Mineshafts	Improved mineshafts

Mod	Role
YUNG's Better Strongholds	Larger, more complex strongholds
YUNG's Better Desert Temples	Multi-room desert temples
YUNG's Better Jungle Temples	Jungle temples with traps
YUNG's Better Ocean Monuments	Redesigned ocean monuments
YUNG's Better Witch Huts	Expanded witch huts
YUNG's Better Nether Fortresses	Improved nether fortresses
YUNG's Better Caves	Larger, more varied caves
Structory	New hand-crafted structures
Towns & Towers	Better village generation

Dimensions

Mod	Role
The Aether	Sky dimension — safe islands, gear progression, dungeons with bosses
Deeper and Darker	The “Otherside” dimension — deep dark themed, warden gear
Twilight Forest	Classic dimension — 6+ bosses with progression gates, unique loot, enchanted forest biomes

End Overhaul

Mod	Role
BetterEnd NeoForge v21.0.25	Full End overhaul — 24+ biomes, new mobs, gear, custom music
Moog's End Structures v2.0.3	New structures to fill the End dimension
Nullscape (data pack)	End terrain overhaul — surreal alien landscapes, new sky colors

Nullscape: A data pack that transforms the End’s terrain into alien shapes (floating islands, spires, arches, voids) with unique sky colors per biome. **Compatible with BetterEnd** — Nullscape’s own documentation states it works with BetterEnd specifically (it is incompatible with most End worldgen mods, but BetterEnd is the documented exception). Nullscape handles terrain shape and sky; BetterEnd handles biomes, mobs, gear, and structures. Install as a data pack in the world’s [datapacks/](#) folder.

New Dependencies (End Overhaul)

Mod	Role
BCLib NeoForge	Core library — required by BetterEnd
WorldWeaver NeoForge	Worldgen library — required by BetterEnd
WunderLib NeoForge	Utility library — required by BetterEnd
Moog's Structure Lib	Library — required by Moog’s End Structures
Cristel Lib	Library — required by Towns & Towers
Blueprint	Library — required by Upgrade Aquatic

Navigation

Mod	Role
Explorer's Compass	Craftable structure/biome finder

Mobility

Mod	Role
Immersive Aircraft 1.1.5	Flyable aircraft — biplane, airship, cargo airship, warship, gyrocopter. Upgradable, colorable
Small Ships 2.0.0-b2.1	Sailable ships — brigantine, cog, drakkar, galley. Build, crew, and navigate; moveable bases on water. Beta (NeoForge)

Configuration — Wave 2

Aether (`aether-common.toml`):

- `"Use default Curios' menu" = true` — Uses Curios API menu instead of Aether's separate accessories screen.
- `"Gives player Aether Portal Frame item" = false` — Player must build the portal manually.
- `"Gives starting loot on entry" = true` — Grants Book of Lore and Golden Parachutes on first entry.
- `"Show Patreon message" = false` — Removes Patreon nag screen.
- `"Add Temporary Freezing automatically" = false` — Disables Aether's temporary freezing datapack.
- `"Add Ruined Portals automatically" = false` — Disables ruined portal datapack.

Interdependency note — Immersive Aircraft + Aether: Aircraft can fly in the Aether dimension. An airship is a natural way to navigate Aether islands — it pairs well with the dimension's sky-island geography.

Interdependency note — Twilight Forest + Apotheosis: Apotheosis boss affixes (Wave 3) apply to Twilight Forest bosses. A Naga with the "Tough" affix is significantly harder. This is intended — it keeps dimension bosses relevant even with late-game gear. If a boss feels too strong, the player can return after better equipment or use Apotheosis gems/runes for the edge.

Interdependency note — Towns & Towers + Cristel Lib: Towns & Towers requires **Cristel Lib** (included in Wave 2 Dependencies above). Without it the game will crash at startup. Cristel Lib provides structure config and runtime datapack support that Towns & Towers relies on for its biome-specific village variants.

Wave 3 — Equipment Magic & RPG

RPG Progression

Mod	Role
Skill Tree (RPG Series)	XP leveling — spend points on attributes and passives
Pufferfish's Skills	Underlying skill system framework

Equipment

Mod	Role
Simply Swords	60+ unique melee weapons with passive/activated abilities
Too Many Bows	30+ unique bows with special abilities and custom attributes
Spartan Weaponry Unofficial	Spears, halberds, pikes, lances, throwing knives — fills the polearm gap
Relics RPG	Legendary accessories with unique abilities
Runes	Socketable runes for weapons/tools
Curios API	Accessory slots (rings, amulets, belts, gloves)

Loot & Affixes

Mod	Role
Apotheosis	Rarity tiers, random affixes, gem socketing, enchanting overhaul. Replaces Savaru's Affixology with more depth

Configuration — Wave 3

Apotheosis — the most config-heavy major mod. Apotheosis is split into modules, each with its own config:

Adventure module — affixes, gems, boss modifiers, rogue spawners:

- Keep all affix types enabled: Ruthless (crit), Arcane (spell), Sturdy (armor), Nimble (speed), Plentiful (mining), Titanic (HP). Each supports a different playstyle.
- Gem tiers enabled: Rough, Cut, Flawless. The gem progression (Rough → Cut → Flawless) provides a satisfying vertical upgrade path. Each tier unlockable through Apotheosis's reforging mechanic.
- Gem types cover all primary stats: melee damage, projectile damage, magic damage, armor, max HP, mining speed, movement speed, luck. This ensures every build archetype benefits from gem-socketed gear.
- Socket count: Gear drops with 1-2 sockets by default. Reforging at the Smithing Table can add a 3rd socket at high rarity (Mythic+). This prevents socket overload while keeping the system meaningful.
- Boss modifiers: all active. Apothic Invaders and Elites spawn naturally with random affixes. They always drop affix items — this is the primary source of mythic gear.
- Rogue Spawners: enabled. Underground spawner+chest combos. Risk vs reward.
- Affix item probability: default rates (~5% base for rare+, scaling with luck and difficulty). Common-enough to feel rewarding, rare-enough to keep each upgrade exciting.

Enchanting module — `config/apotheosis/enchanting.cfg`:

- Keep all enchantment overhauls enabled: higher level cap (up to level 150+), tiered bookshelves (Hell, Seashelf, Endshelf), Library of Alexandria, new enchantments (Chromatic, Mounting, Stable Footing, etc.).
- The Library of Alexandria stores all enchanted books in one block — essential for enchanting organization at scale.
- Enchantment max levels increased for many vanilla enchantments: Protection can reach level 8, Sharpness level 6, Fortune level 5, Looting level 5, etc. These higher caps match the power fantasy curve — gear enchanted at max level should feel mythic.

Potion module:

- Potion Charms enabled: craft a charm from blaze powder + potions, equip in offhand for passive effect. Reduces inventory clutter from potion bottles.
- Potion augmentation allowed: glowing, brewing, changing base type into splash/lingering at standard ratios. Augmented potions can have higher level caps (e.g., Potion of Strength III).

Spawner module:

- Spawner capture and modification enabled. Silk-touch a spawner with a pickaxe, then modify it (entity type, spawn delay, player range, min/max entities). Key for XP farms and mob-based automation.
- Spawner stats are fully configurable after placement through the spawner's GUI (right-click with an empty hand). Players can fine-tune delay, count, range, and conditions per spawner — no external tool needed.

Village module:

- Wandering trader improvements enabled. Traders can sell Apotheosis items (gems, affix gear, enchanted books) at elevated prices. Adds a secondary acquisition path for players who prefer trading over combat.

Deadly module: Enabled. Apotheosis's Deadly module adds randomly generated death-traps to the world (sudden spikes, poison darts, instakill rooms, lava traps). This adds risk-reward tension to structure exploration — the deadliest rooms hold the best loot. The Deadly module's traps are telegraphed and avoidable with caution, and GraveStone Mod keeps your items safe if you do trigger one, so the chill rule is preserved.

Reforging costs — Apotheosis's reforging mechanic (reroll affixes with gem dust) uses default material costs. As the player progresses, reforging costs scale:

- Common→Uncommon: cheap (a few gem dust)
- Rare→Epic: moderate (dust + some ingots)
- Epic→Mythic: expensive (dust + rare materials like netherite scrap/echo shards)

This cost curve encourages the player to settle for “good enough” gear until they're ready to invest in perfect rolls.

New Dependencies (Wave 3)

Mod	Role
Placebo	Core library — required by Apotheosis
Apothic Attributes	Attribute system module — required by Apotheosis
Apothic-Enchanting	Enchanting module — required by Apotheosis
Apothic-Spawners	Spawner module — required by Apotheosis
Patchouli	Guidebook library — required by Apotheosis

Mod	Role
Simply Tooltips	Tooltip framework — required by Simply Swords
Fuzzy Config	Config library — required by Simply Swords
Ranged Weapon API	Ranged combat API — required by Relics RPG and Skill Tree
Spell Engine	Magic framework — required by Relics RPG and Skill Tree
Bundle API	Bundle/container API — required by Runes

Interdependency notes for Wave 3

- **Spell Engine** is required by both Relics RPG and Skill Tree (RPG Series) — it was previously considered for removal but must be retained as a mandatory dependency of these core RPG mods.
 - **Ranged Weapon API** is shared between Relics RPG and Skill Tree — install once, used by both.
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Wave 4 — Food, Farming & Comfort

Mod	Role
Farmer's Delight 1.3.2	Cooking hub — cutting board, cooking pot, stove, new crops
Brewin' And Chewin' 4.5.0	Drinks and fermentation — beer, wine, mead, whiskey
Storage Delight 1.3.0	Kitchen furniture — drawers, cabinets, shelves for ingredient storage
More Delight 26.04.18a+1.21-neoforge	Extra meals, sandwiches, toasts, and knife variants
My Nether's Delight 1.10.2	Nether-themed food ingredients and recipes
Ocean's Delight 1.0.4	Ocean-themed seafood dishes
Ender's Delight v1.3.0	End-themed food for Farmer's Delight (moved from Wave 2)
Neo Bee Fix	Fixes bee AI bugs — bees leave hives, don't fall from sky, no NW wander bias

New Dependencies (Wave 4)

Mod	Role
Delight Lib 26.05.18-1.21-neoforge	Library — required by More Delight v26.04.30+ for cross-loader addon dev

Interdependency notes for Wave 4:

- Farmer's Delight meals provide status effects that stack with Brewin' And Chewin' drinks. Combined, a full meal + a vintage drink gives significant combat buffs — useful before boss fights (Wave 5).
- AE2's ME System can store Farmer's Delight ingredients and cooked meals. Set up an Export Bus on a cooking pot for automated ingredient delivery.
- Bountiful (Wave 4.5) can generate bounties that request Farmer's Delight meals as objectives, and reward cooking ingredients.
- My Nether's Delight, Ocean's Delight, and Ender's Delight unlock food sources from cross-dimensional ingredients (Nether, Overworld oceans, and End), giving reasons to explore every dimension beyond just combat.
- Storage Delight cabinets and drawers provide early-game kitchen storage that feeds into AE2 in Phase 2. They're designed as stepping stones, not replacements.
- More Delight v26.04.30+ requires **Delight Lib** as a hard dependency — install both or neither.
- **Serene Seasons** (Wave 2) affects crop growth rates seasonally — crops planted out-of-season grow slower. This adds depth to greenhouse planning with Farmer's Delight crops. Temperature damage is disabled, so crops are the only seasonal pressure.

Wave 4.5 — Quests

Procedural bounty system that gives purpose to exploration and material collection without manual quest authoring.

Mod	Role
Bountiful 8.0.0-beta.2	Bounty boards in villages — procedurally generated fetch/kill/exploration quests with rewards

Bountiful generates quests automatically — find a bounty board in any village, accept tasks, earn rewards. Fully optional. Items from installed mods automatically appear in its reward pools.

Wave 5 — Combat & Mobs

Mod	Role
Better Combat	Melee combat animations — weapon reach, sweeping, attack speed
Mutant Monsters	Mutant zombie, skeleton, creeper, enderman — unique drops
L_Ender's Cataclysm	Epic boss fights — Leviathan, Netherite Monstrosity, Ender Guardian (CurseForge, NeoForge 1.21.1)
When Dungeons Arise	Roguelike combat structures with loot
Enchantment Descriptions	Shows enchantment effects in tooltips
Create Big Cannons	Artillery engineering — siege cannons for colony defense and spectacle
Cut Through	Combat QoL — swing through transparent blocks (tall grass, cobwebs) without breaking them

Apotheosis also adds boss modifiers and elite mob affixes — partially covers what Champions would have done.

New Dependencies (Combat)

Mod	Role
Ritchie's Projectile Library	Library — custom projectiles and trajectories (Create Big Cannons)
playerAnimator	Animation library — required by Better Combat
Puzzles Lib	Library — required by Mutant Monsters
Citadel	Library — required by L_Ender's Cataclysm
Lionfish-API	Library — required by L_Ender's Cataclysm
Bookshelf	Library — required by Enchantment Descriptions

Better Combat + Simply Swords: Simply Swords weapons have built-in Better Combat data files — each weapon category uses the correct attack animation, reach, and speed automatically.

Cut Through: Works alongside Better Combat — lets you swing through transparent blocks (tall grass, vines, cobwebs) without breaking them, keeping your attack chain uninterrupted. Client-side only, no conflicts with any combat mod.

Wave 6 — Building & Decoration

Core Decoration

Mod	Role
Rechiseled	600+ decorative block variants with connected textures
Supplementaries	Vanilla+ decoration — jars, signposts, faucets, weather vanes
Macaw's Bridges	Bridges in all wood types
Macaw's Doors	Vanilla-style doors for every wood
Macaw's Roofs	Roofing blocks
Macaw's Windows	Window variants
Building Wands	Faster large-scale construction
Handcrafted	Furniture — chairs, tables, shelves, curtains
The Magic Mirror	Reflective mirror block that shows the nearest player's reflection

Rechiseled Addons

Mod	Role
Rechiseled: Create	100+ decorative variants of Create blocks + mechanical chisel
Rechiseled: Applied Energistics 2	AE2 autocrafting for Rechiseled blocks via Chiseling Pattern Encoder

New Dependencies (Wave 6)

Mod	Role
Moonlight Library	Core library — required by Supplementaries
Resourceful Lib	Core library — required by Handcrafted
Fusion (Connected Textures)	Connected textures — required by Rechiseled

Wave 7 — Space Exploration

Core Space Mod

Mod	Role
Stellaris 1.4.23	Space exploration — rockets, Moon/Mars/Venus/Mercury/Glacio, oxygen system, machines, energy

Stellaris (by ExodusTeam) is the successor to Beyond Earth and the premier space mod for NeoForge 1.21.1. It features 4 rocket tiers, a rover, oxygen management, custom gravity per planet, oil/water/fuel processing, radioactives, and alien mobs. Machines include the Oxygen Distributor, Water Separator, Solar Panel, Fuel Refinery, and Pump Jack. 250+ building blocks across planet-specific sets.

Dependencies

Mod	Role
Architecture API	Cross-loader library (already in Wave 0)
Potentials API	Required library (Stellaris dependency)

Sky Aesthetics is embedded in Stellaris — no separate install needed.

Integration with Tech Mods

Layer	Stellaris ↔ Tech Mod Interaction
Power	Mekanism cables can power Stellaris machines. Create rotational → Mekanism → Stellaris
Energy	Mekanism universal cable connects to Stellaris machines (energy API compat)
Storage	AE2 ME drives store Stellaris materials; autocrafting builds rocket components
Transport	Create trains move space resources from landing pad to base. Sophisticated Backpacks carry oxygen tanks
Ore Processing	Mekanism 5x ore multiplication works on Stellaris ores (desh, ostranium, etc.)
Food	Farmer's Delight meals eatable in space when oxygen is present (v1.2.1+)
Gear	Apotheosis affixes may apply to Stellaris space suit (if item tags align)

Progression Placement

Phase 2 — Industrial Ascent (hours 40–200). Rockets require advanced materials that gate them behind Mekanism mid-game processing and AE2 infrastructure. Oxygen systems and fuel refining add a parallel supply-chain puzzle that rewards industrial automation.

Mod Count Summary

Wave	Mods	Deps	Total	Notes
Wave 0 — Foundation	58	9	67	Performance, QoL, time control, storage, travel, graves, village trading, UI, chunk loading (+6 resource packs, 1 shaderpack, 1 data pack)
Wave 1 — Tech	18	4	22	Create + 12 addons (incl. Power Grid, Gunsmithing, Ornithopter Glider, Train Utilities, Railways Navigator, Interiors, Train Parts, Threaded Trains), Mekanism + Generators, AE2, Advanced Finders, Advanced Chimneys + NTGL, GeckoLib, Framework, ForgeEndertech deps
Wave 1.5 — Colony	5	—	5	MineColonies (4 deps counted as mods — they're library mods; CurseForge-only)
Wave 2 — Exploration	24	6	30	YUNG's (9), Terralith/Tectonic, Serene Seasons, Darker Depths, Upgrade Aquatic, dimensions (3), End overhaul (2 + Nullscape dp + 6 deps), navigation, aircraft, ships
Wave 3 — Equipment Magic	9	10	19	Skill Tree, Simply Swords, Too Many Bows, Spartan Weaponry, Relics, Runes, Curios, Apotheosis + 10 deps (Placebo, Apothic modules, Patchouli, Simply Tooltips, Fzzy Config, Ranged Weapon API, Spell Engine, Bundle API)
Wave 4 — Food & Farming	8	—	8	Farmer's Delight + 6 addon mods + Neo Bee Fix
Wave 4.5 — Quests	1	—	1	Bountiful
Wave 5 — Combat	7	6	13	Better Combat, Mutant Monsters, Cataclysm, Dungeons Arise, Big Cannons, Cut Through, Enchantment Descriptions + RPL lib, playerAnimator, Puzzles Lib, Citadel, Lionfish-API, Bookshelf
Wave 6 — Building	10	3	13	Rechiseled, Supplementaries, Macaw's (4), Building Wands, Handcrafted + Rechiseled: Create, Rechiseled: AE2 + Moonlight Lib, Resourceful Lib, Fusion
Wave 7 — Space Exploration	1	1	2	Stellaris, Potentials API
Total	140	40	180	All confirmed NeoForge 1.21.1